

Algorithmic Market Making Capstone Case

Rotman International Trading Competition (RITC)x

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<https://www.rotman.utoronto.ca/faculty-and-research/education-labs/bmo-financial-group-finance-research-and-trading-lab/rit-market-simulator>

OVERVIEW

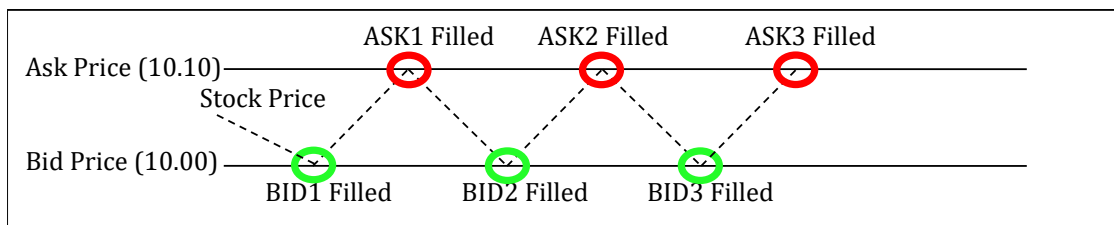
The Algorithmic Market Making Capstone Case extends the original single-security market-making setup into a multi-security environment with more realistic fee–rebate (“feebate”) structures. Participants design and implement algorithms that make markets across several securities while adapting to changing spreads, liquidity, and order-flow dynamics.

The case emphasizes the trade-off between capturing bid–ask spreads and exchange rebates and controlling inventory and market risk, under practical constraints such as position limits and order-rate caps.

Participants should develop a market making algorithm that is able to pursue the most profitable opportunities across securities and over time in an environment in which the potential source of returns from earning the bid–ask spread (versus focusing just on rebates) is enhanced.

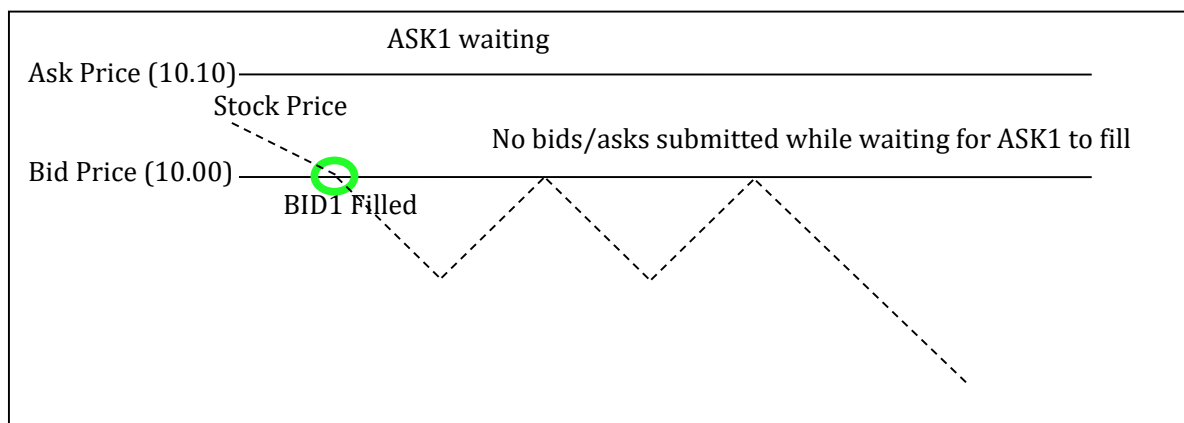
DESCRIPTION

Trading occurs over time. Capturing a bid–ask spread will require participants to inventory a long (or short) position for an uncertain period of time as they wait for an opposing trade to cover their position. The simplest execution of a market-making strategy is to submit a paired bid and offer and have the two orders filled over time; the participant (algorithm) earns the price differential. When markets aren’t trending, this is a reasonably effective strategy.



The participant submits their first “pair” of orders (BID₁ and ASK₁) of equal size and waits for the two to become filled. The price trades down to the bid price, which fills, and the participant has an inventory of shares. The participant then waits, and the price trades up to the ask price and they receive a fill at the ask price, extinguishing their inventory. They no longer have a position, and they have successfully captured a 10 cent bid/ask spread. The participant can then submit a new pair of orders and repeat the process.

This logic fails in a market that does not stay in a small trading range, because if one of the orders does not get filled, the algorithm waits (potentially) a very long time, and fails to capture the bid ask spread during that time.



The logical answer to this is that if a pair order does not get filled on both sides (that is, either the bid or the ask order does not get filled), one would want to simply “reset” the algorithm and start submitting new sets of pairs. As a starting strategy, we can simplify this into two simple rules:

1. We always want to have a bid and an ask order in the market. For example, the bid could be submitted at (LAST – Spread) and the ask submitted at (LAST + Spread). We want our pair of orders to be close to the top of the book.
2. We want our position to be close-to-balanced whenever possible. When markets are trending and we’re getting filled on more bids than asks, we want to adjust our algorithm accordingly*.

Programmatically, your algorithm should do the following separately for each security you trade:

1. Constantly check to see if you have any orders in the order book. If you do not, then you should submit your bid and ask. If you only have one order in the order book, cancel it.
2. Check your inventory and adjust your bid/ask prices or quantities to try to balance your inventory.

*Think about how you would adjust your algorithm “accordingly” to execute the trades and attempt to balance your inventory.

MARKET DYNAMICS

In this case, we introduce multiple securities, which allows participants to develop strategies that are different across securities with different characteristics.

The table below provides a description of the securities.

Security	Active Trade Fee (\$/share)	Passive Trade Rebate (\$/share)
CNR	0.0027	0.0023
RY	-0.0014	-0.0020
AC	0.0015	0.0011

Fees and rebates are applied to all trades based on whether they remove liquidity (active orders) or provide liquidity (passive orders). Market orders or the portion of a limit order that crosses the market immediately remove liquidity. Limit orders (including the portion of a limit order that is not marketable, that is, not immediately executed) provide liquidity. The feebate structures are calibrated to approximate that in a traditional exchange such as the TSX for a cross-listed stock (the CNR security) with a fee charged for removing liquidity and a rebate for providing liquidity, a newer competing exchange such as NASDAQ's CX2 (the RY security) that *pays a rebate* for removing liquidity and *charges a fee* for providing liquidity (represented by the negative signs in front of the fee/rebate amount), and a traditional exchange such as the TSX with a non-cross-listed stock (the AC security) that provides a lower passive trade rebate in the absence of competition.

For example, assume that the top of the limit-order book for AC is currently:

Bid Size	Bid Price	Ask Price	Ask Size
1000	9.99	10.02	1000
1000	9.98	10.05	2000

If a market-making participant submits a limit order to buy 1,500 shares with a limit price of 10.03, s/he will immediately receive 1,000 shares for a price of \$10.02 and will be charged \$1.50 [= \$0.0015 × 1,000] in fees. The remaining 500 shares will stay in the limit order book (as a bid for 500 shares at \$10.03) and will receive a rebate of \$0.55 [= \$0.0011 × 500] if/when filled.

The markets will be open for 5 minutes (300 seconds) for each replication of the case. As a result of the differences in fees and rebates outlined in the table above, participants are expected to apply market-making strategies across the three different securities. To further challenge strategies and to put emphasis on focusing on earning the bid-ask spread as well as potential rebates, the dynamics of each available security are buffeted by differential liquidity from the programmed order flow (the ANON liquidity traders) such that the dynamics of the available spread will vary across securities and over time.

As noted above, market-making strategies may result in accumulated inventory. As a result, they will be exposed to market risk. There is a limit of 5,000 shares per order, and each participant is limited to a position of 25,000 shares gross or net. In order to discourage excessive exposure to market risk, trades that would cause a violation of either limit will be rejected.

Participants can use the REST API. The maximum orders per second have been set at 500.

KEY OBJECTIVES

Objective 1

Decide what the appropriate bid/ask spread should be when you submit your pair orders, taking market conditions, trading fees and rebates into account.

Objective 2

Identify the different ways you can alter your trading strategy, so that you keep submitting orders but attempt to balance your book.

Objective 3

Assess how market trends and correlated price movements affect market-making strategies across the three securities.

Objective 4

Pursue the most profitable opportunities across securities and over time by adapting strategies to each security's characteristics.